

Discover more

K-Map practice problems

Computer science textbook

maps

Karnaugh Map

Number of Variables:

4 Variables

Clear All

Fill All with 1

CD
AB

00011110

X11X

0132

1001

4576

0101X

12131514

1010

891110

Click cells to toggle values: 0 → 1 → X → 0

X represents "don't care" condition (can be either 0 or 1)

Simplified K-Map Result

SOP Form

POS Form

CD
AB

00011110

X11X

0132

1001

4576

0101X

12131514

1010

891110

Simplified Boolean Expression

A'D' + AD + B'C' + B'D

Original Groups

Original

A'B'C'D' + A'B'CD' + A'BC'D' + A'BCD'

Minterms

(0, 2, 4, 6)

Simplified

A'D'

Original

AB'C'D + AB'CD + ABC'D + ABCD

Minterms

(9, 11, 13, 15)

Simplified

AD

Original

A'B'C'D' + A'B'C'D + AB'C'D' + AB'C'D

Minterms

(0, 1, 8, 9)

Simplified

B'C'

Original

A'B'C'D + A'B'CD + AB'C'D + AB'CD

Minterms

(1, 3, 9, 11)

Simplified

B'D

Logic Circuit

Truth Table

#	A	B	C	D	Output		
0	0	0	0	0	0	1	X
1	0	0	0	1	0	1	X
2	0	0	1	0	0	1	X
3	0	0	1	1	0	1	X
4	0	1	0	0	0	1	X
5	0	1	0	1	0	1	X
6	0	1	1	0	0	1	X
7	0	1	1	1	0	1	X
8	1	0	0	0	0	1	X
9	1	0	0	1	0	1	X
10	1	0	1	0	0	1	X
11	1	0	1	1	0	1	X
12	1	1	0	0	0	1	X
13	1	1	0	1	0	1	X
14	1	1	1	0	0	1	X
15	1	1	1	1	0	1	X

Click buttons to set output values: 0 or 1 or X

X represents "don't care" condition (can be either 0 or 1)

Minterms

Maxterms

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